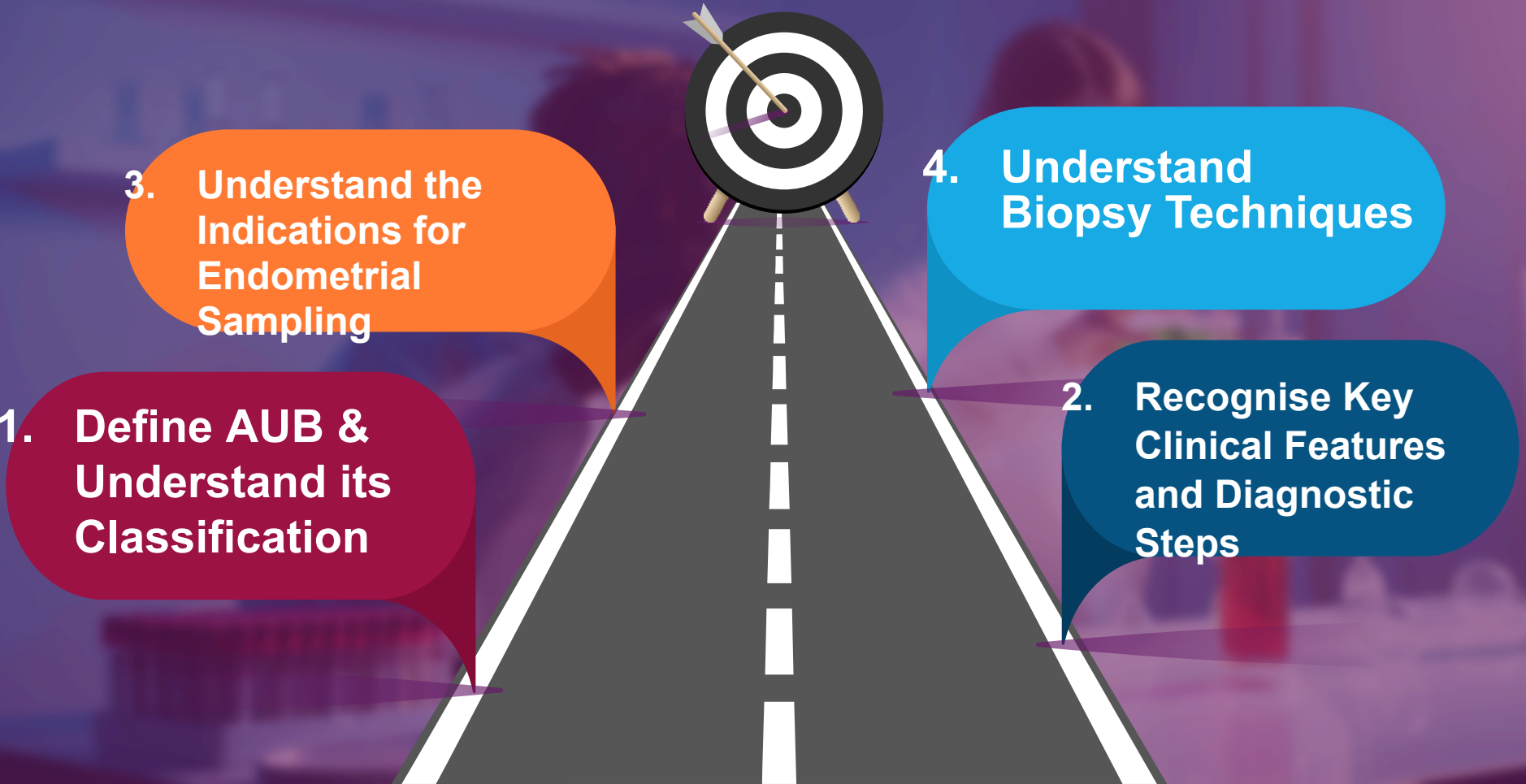


Abnormal Uterine Bleeding and Endometrial Sampling

Prepared By: Aro Dana, Lava Wrya, Kizhe Kawa, Rozhi Othman

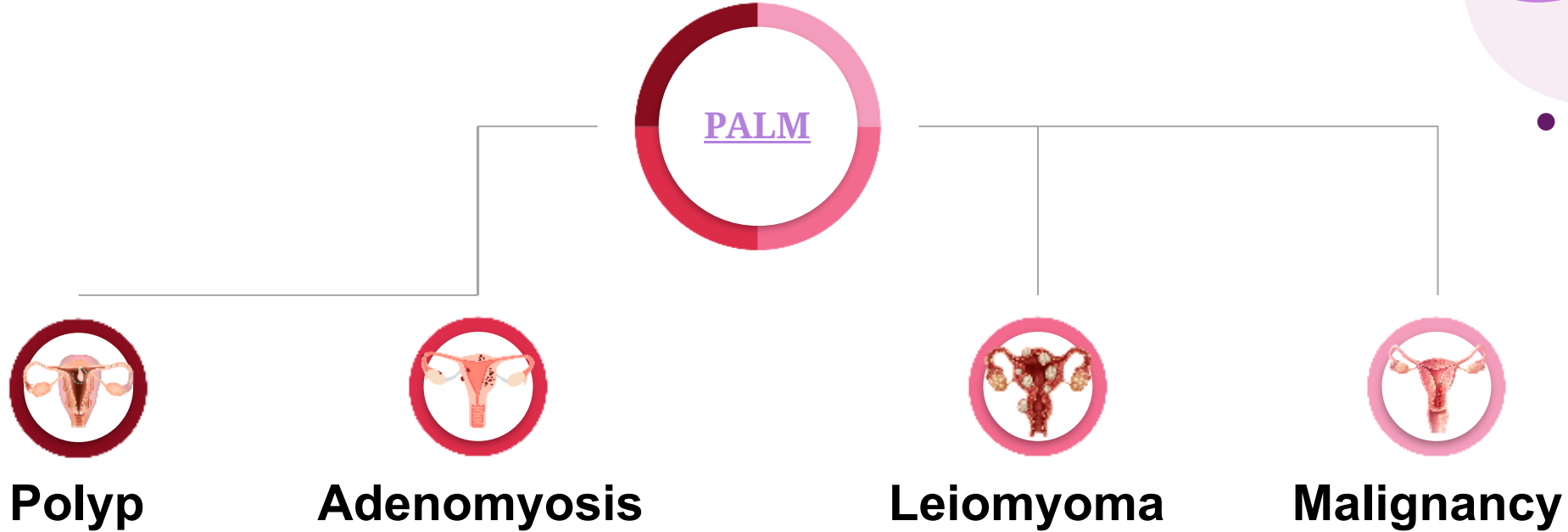
Objectives of This Seminar Are:



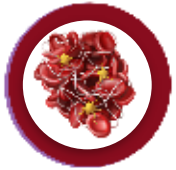
• What is Abnormal Uterine Bleeding (AUB)? •

- AUB: Is a broad term for uterine bleeding that's abnormal in its: Frequency, Regularity, Duration, or Volume.
 - It replaces older terms like menorrhagia or metrorrhagia.
 - AUB is a Symptom and NOT a Diagnosis.
 - Classified by FIGO system to Structural and Non-Structural causes.
 - AUB affects up to 30% of women in their reproductive age.

Structural Causes



Non-Structural Causes



Coagulopathy



**Ovulatory
Dysfunction**



Endometrial



Iatrogenic



**Not
Classified**

Patient Presentation

- **Heavy Menstrual Bleeding (HMB):** Excessive menstrual blood loss which interferes with a woman's life in Physical, Emotional, Social and Material aspects of life.
- **Intermenstrual Bleeding (IMB):** Bleeding between expected cycles.
- **Post-Coital Bleeding (PCB):** Bleeding after intercourse.
- **Post-Menopausal Bleeding (PMB):** Bleeding >12 months after the last menstrual period.





Clinical Evaluation - History

- **Important Points to be asked while taking History:**

- Pregnancy Status: Always Exclude !
- Nature of Bleeding: Frequency, Duration, Volume, Intermenstrual or Postcoital Bleeding
- Impact: How does it affect her QoL?
- Associated Symptoms: Pain, Pressure, Discharge, Infertility
- Medical History: PCOS, Thyroid, Bleeding Disorders.
- Drug History: Tamoxifen, Anticoagulants, Contraception
- Cervical Screening: Is it up to date?

• Clinical Evaluation - Examination •

- Vital Signs
- Inspection
- Abdominal Palpation
- Speculum Examination
- Bimanual Examination

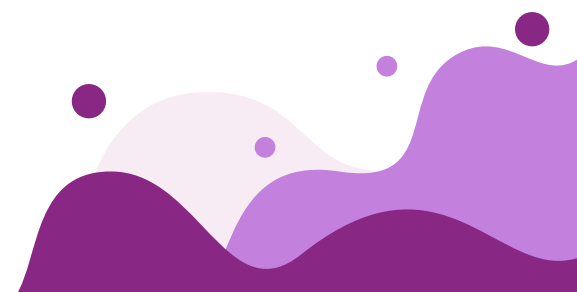
Initial Investigations



- ❑ Pregnancy Test
- ❑ Full Blood Count
- ❑ Coagulation Profile Only if HMB started at menarche or there's a family/personal history for clotting disorders
- ❑ Do NOT do routine hormonal (FSH, LH, etc.) or thyroid tests unless there are other symptoms of thyroid disease or PCOS (e.g., irregular cycles, hirsutism).

Imaging

- ❖ Transvaginal Ultrasound (TVUS) is the first-line imaging for AUB if:
 - The uterus is palpably enlarged.
 - The history or exam suggests a structural abnormality (PALM).
- ❖ Outpatient Hysteroscopy if intrauterine pathology is suspected.
- ❖ MRI if ultrasound was inconclusive.

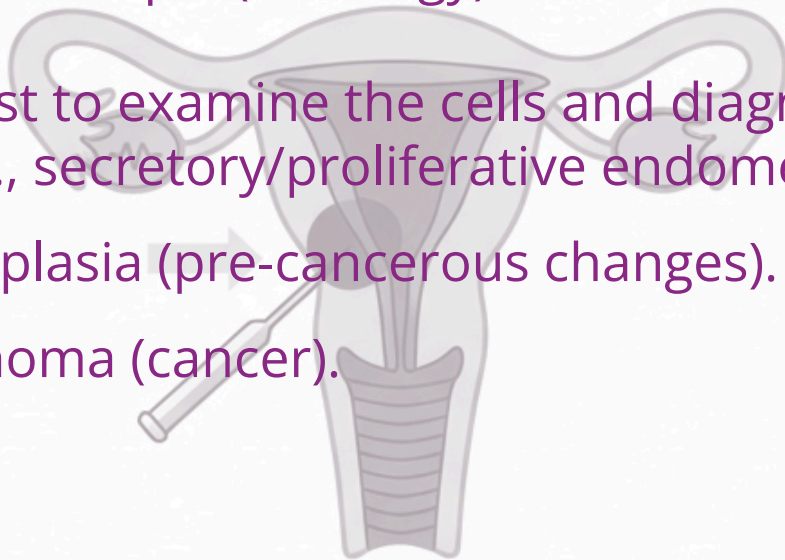


Endometrial Sampling

It is to obtain a tissue sample (histology) from the endometrial lining.

To allow a pathologist to examine the cells and diagnose:

- Benign finding (e.g., secretory/proliferative endometrium, polyp).
- Endometrial Hyperplasia (pre-cancerous changes).
- Endometrial Carcinoma (cancer).





Indications for Endometrial Sampling

- Any PMB.

- Persistent IMB.

- HMB Unresponsive to Medical Treatment.

- Suspicious TVUS Findings:

 - Thickened Endometrium $>4\text{mm}$.

 - Irregular Endometrial lining.

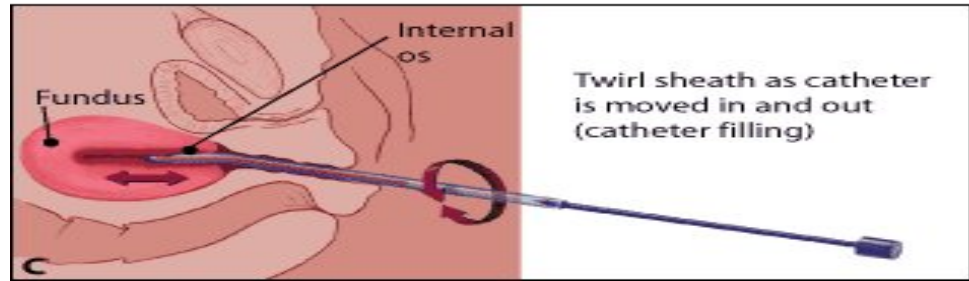
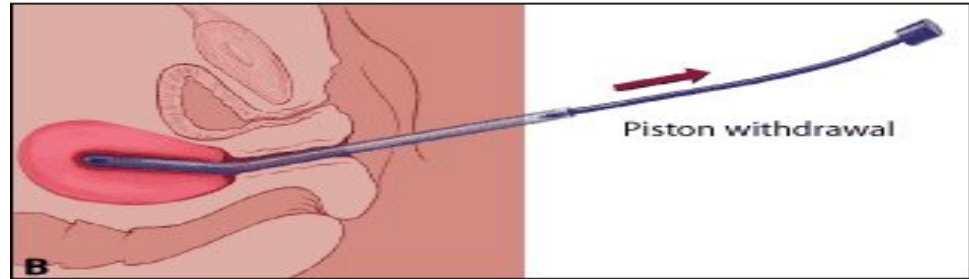
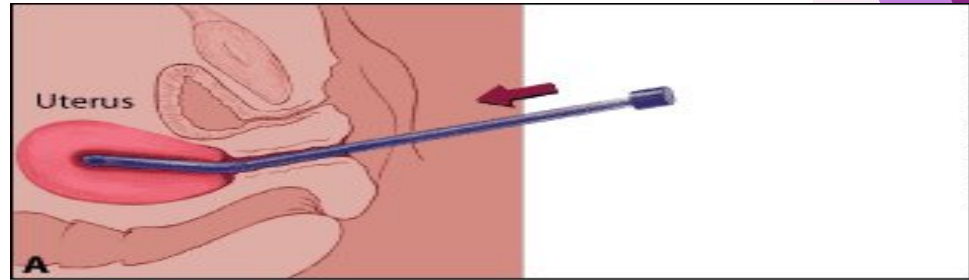
 - Polypoid Mass.

- High-Risk Patients, e.g. : Tamoxifen, Obesity, Nulliparity, Lynch syndrome, PCOS.

Pipelle Endometrial Biopsy

Procedure:

- Done in an outpatient clinic,
- No general anaesthesia (GA).



Advantages

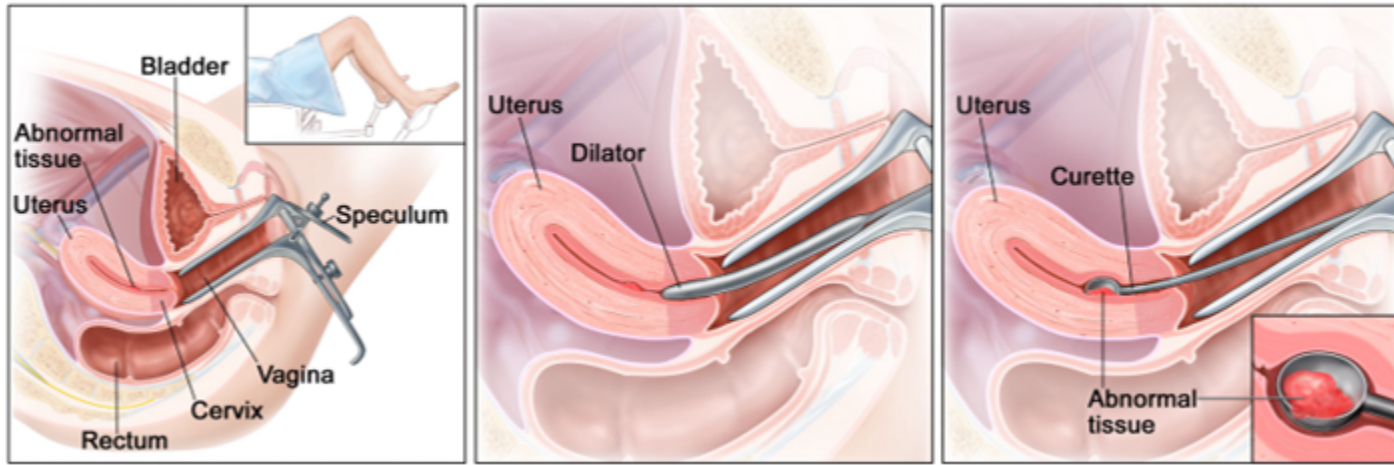
- High diagnostic accuracy for diffuse cancer (>98%).
- Quick, low-cost, avoids GA.

Disadvantages

- Can be painful.
- Risk of "inadequate sample", especially in post-menopausal women.
- It's a "blind" sample - can miss focal lesions.
- Not recommended by "NICE"

Dilation & Curettage (D&C)

“Old method”: Dilates cervix and uses a metal curette to scrape the endometrium.



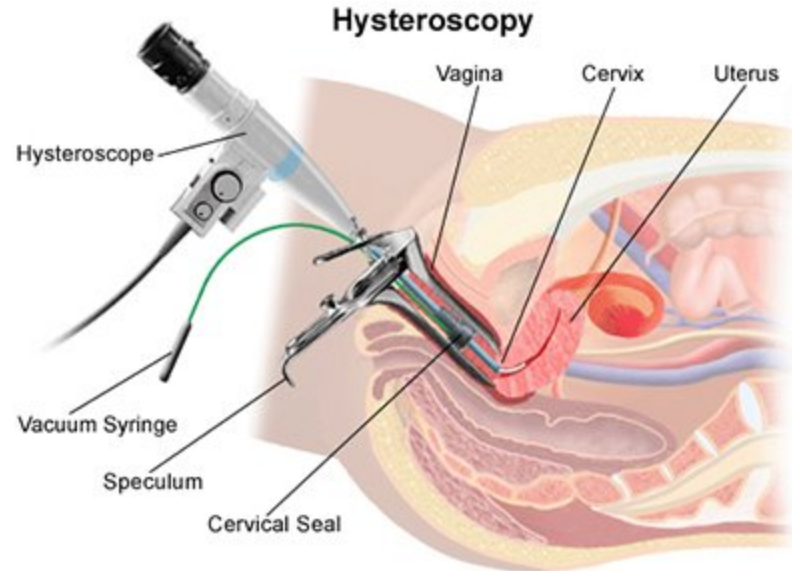
D&C is still used therapeutically (e.g., removing retained products of conception), but not for diagnosing AUB.

Hysteroscopy + Targeted Biopsy (Gold Standard)

Procedure:

A thin, lighted camera (hysteroscope) is passed through the cervix into the uterine cavity.
The cavity is distended (usually with saline).
Allows direct visualization of the entire endometrial lining.

□ "See-and-Treat".

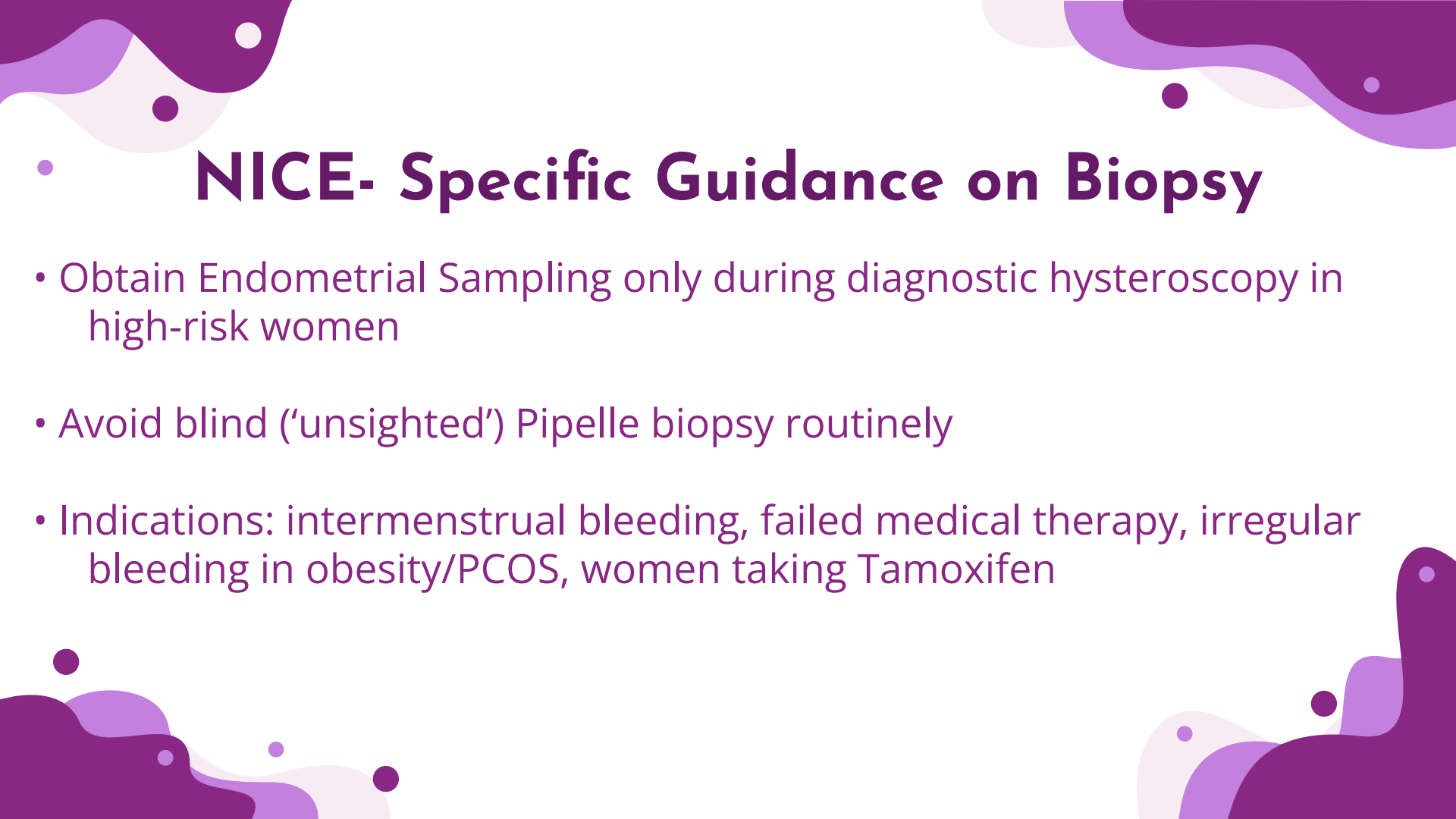


Advantages

- “Gold Standard”: Can identify and target focal lesions missed by a Pipelle.
- Can be diagnostic and therapeutic.

Disadvantages

- More invasive, more discomfort, requires specialized equipment and clinic.



• **NICE- Specific Guidance on Biopsy**

- Obtain Endometrial Sampling only during diagnostic hysteroscopy in high-risk women
- Avoid blind ('unsighted') Pipelle biopsy routinely
- Indications: intermenstrual bleeding, failed medical therapy, irregular bleeding in obesity/PCOS, women taking Tamoxifen

Management

Management of patients with AUB depends on Age, Risk, Severity and Response to treatment.

If they had HMB, no Red Flags, Normal TVUS.

- ❖ Our Primary Goal will be Managing Symptoms:
 - First-Line: Levonorgestrel-IUS (Mirena)
 - Alternative Treatments: Tranexamic Acid, NSAIDs, COCP, POP.
- ❖ Refer to Gynaecologist if medical management fails or is not suitable.



Management

If Structural Causes were found on TVUS,
Our Primary Goal is to Manage Symptoms AND The Structural problems.

Management Options:

- Medical Management, to see if they control the bleeding.
- Refer to a Gynaecologist to discuss surgical options:
 - Hysteroscopy
 - Uterine Artery Embolization (UAE)
 - Myomectomy
 - Hysterectomy
 - Second Generation Endometrial Ablation



Management

- If we had a “Suspicious” Patient (PMB or IMB), we perform Endometrial Sampling and Act based on the Histology report.
 1. Benign (e.g. Normal, Polyp):
Reassure the Patient and Manage the Symptoms accordingly.
 2. Inadequate Sample:
Repeat the sample, if Pipelle was done try to switch to Hysteroscopy.
 3. Hyperplasia without Atypia:
This is a Low-risk Pre-cancerous stage.
Manage with Mirena IUS
Follow-up the patient and repeat the biopsy in 6 months



Management

4. Atypical Hyperplasia (Pre-Cancer):
This has high risk of progression.
If she had completed her family, refer for Total Hysterectomy.
If fertility is desired, give High-dose Progestogens and follow-up.
5. Malignancy (Endometrial Cancer)
Confirm the diagnosis.
Refer to Gynaecology-Oncology Team for staging and definitive treatment.



Case 1

- Patient: 44-year-old woman, HMB affecting her job. No IMB. Examination reveals: Uterus normal size. Speculum normal. Investigations: FBC: Hb 10.5 g/dL (anaemic). TVUS: Normal uterus, 5mm regular endometrium.
 - Does she need a biopsy?

No. According to NICE, she has no red flags (like IMB or PMB) and a non-suspicious TVUS. We treat her symptoms.

NICE Management (NG88):

- 1st Line: Levonorgestrel-IUS (Mirena coil).
- Alternatives: Tranexamic acid, Mefenamic acid, or COCP, POP



Case 2

- Patient: 62-year-old woman, 8 years post-menopause, presents with one episode of vaginal bleeding (PMB). • Exam: Unremarkable.
- Investigations:
 - TVUS: Endometrial thickness 9mm. (NICE "red flag" is >4-5mm for PMB).
 - Does she need a biopsy?
- YES. Absolutely. This is the classic indication.
- Procedure: Given the thickened endometrium, a "blind" Pipelle has a high risk of "inadequate sample" or missing a focal lesion.
- Best Practice: Proceed directly to Hysteroscopy + Biopsy.
- Histology: "Atypical endometrial hyperplasia."
- Management:???

Take-Home Message

- •AUB is a symptom with a wide range of causes (PALM-COEIN).
- NICE Guidelines focus on Quality of Life and a structured, logical pathway.
- Endometrial Sampling is NOT for every patient with AUB.
- Endometrial Sampling is critical for malignancy exclusion
- Sampling is a targeted investigation driven by specific risk factors: PMB, Persistent IMB, and Suspicious Ultrasound Findings.
- Hysteroscopy with targeted biopsy is the gold standard for diagnosis, as it can see and sample focal lesions.

References

- NICE Guideline [NG88] (2018). Heavy Menstrual Bleeding: assessment and management.
- NICE Guideline [NG12] (2021). Endometrial cancer: diagnosis and management.
- RCOG: Management of Endometrial Hyperplasia (Green-top 67)
- FIGO Working Group (2011). FIGO system for nomenclature and classification of causes of abnormal uterine bleeding.





THANKS

